

Practical 7: Shared Preferences

1. Create a new Flutter project type **Application** name **user_profile**. Set the package name to **my.edu.tarc**. This app utilises the **shared_preferences** library to store key-value pairs in local storage. For this case, we will save name and email address of a user as shown below:



2. Open a new **Terminal** and enter the following command:

```
flutter pub add shared_preferences
```

3. Open the **main.dart** file and import the **shared_preferences.dart** file:

```
//TODO 1: - Import reference to async and shared_preferences
import 'dart:async';
import 'package:shared_preferences/shared_preferences.dart';
```

4. In the class **_MyHomePageState** extends **State<MyHomePage>**, insert **TextEditingController** for the two **TextField** widgets in the class:

```
//TODO 2: - Insert TextEditingController
final _nameTextEditingController = TextEditingController();
final _emailTextEditingController = TextEditingController();
```

5. Within the class **SharedPreferencesDemoState** extends **State<SharedPreferencesDemo>**, insert to asynchronous functions as follows: -

```
//TODO 3: - Insert _loadProfile() and _updateProfile() functions
Future<void> _loadProfile() async{
  final prefs = await SharedPreferences.getInstance();
  _nameTextEditingController.text = prefs.getString('name') ?? "";
  _emailTextEditingController.text = prefs.getString('email') ?? "";
}
```

```
Future<void> _updateProfile() async{
  final prefs = await SharedPreferences.getInstance();
  prefs.setString('name', _nameTextEditingController.text);
  prefs.setString('email', _emailTextEditingController.text);
}
```

6. Override the `initState()` and `dispose()` functions as follows:

```
//TODO 4:- Override initState() and dispose() functions
@override
void initState() {
  _loadProfile();
  super.initState();
}

@override
void dispose() {
  super.dispose();
  _nameTextEditingController.dispose();
  _emailTextEditingController.dispose();
}
```

7. In the **body** of the **build()** function, create two **TextField** widgets and a **ElevatedButton** widget.

```
//TODO 5: - Insert TextField and ElevatedButton
TextField(
  controller: _nameTextEditingController,
  keyboardType: TextInputType.name,
  decoration: const InputDecoration(
    labelText: 'Name',
  ),
),
const SizedBox(),
TextField(
  controller: _emailTextEditingController,
  keyboardType: TextInputType.emailAddress,
  decoration: const InputDecoration(
    labelText: 'Email',
  ),
),
const Expanded(child: SizedBox()),
ElevatedButton(
  onPressed: (){
    _updateProfile();
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(
        content: Text('Profile info saved.'),
      ),
    );
  },
  child: const Text('Save')
),
```

8. Run the app, insert a name and an email, press the **Save** button.
9. On the main menu, select **View** -> **Tool Windows** -> **Device Explorer**. In the Device Explorer, click the **data** -> **data** folder.

Name	Permiss...	Date	Size
✓ /	drwxr-xr-x	2009-01-01 00:	4 KB
> / acct	drwxr-xr-x	2009-01-01 00:	4 KB
> / apex	drwxr-xr-x	2024-12-30 09:	1.2 Ki
> / bin	lrw-r--r--	2009-01-01 00:	11 B
> / cache	drwxrwx--	2009-01-01 00:	4 KB
> / config	drwxr-xr-x	2024-12-30 09:	0 B
> / d	lrw-r--r--	2009-01-01 00:	17 B
✓ / data	drwxrwx--	2024-12-30 09:	4 KB
> / data / app	drwxrwx--	2024-12-30 09:	4 KB
✓ / data / data	drwxrwx--	2024-12-30 09:	4 KB
> / data / data / android	drwxrwx--	2024-12-30 09:	4 KB
> / data / data / android.auto_gen	drwxrwx--	2024-12-30 09:	4 KB
> / data / data / android.auto_gen	drwxrwx--	2024-12-30 09:	4 KB

10. Scroll down and look for your application package name. Inside the **shared_prefs** folder, you will find a shared preference file titled **FlutterSharedPreferences.xml**.

✓ / com.example.demo_shai	drwxrwx--x
✓ / com.example.demo_shai / app_flutter	drwxrwx--x
> / com.example.demo_shai / app_flutter / flutter_assets	drwx-----
≡ / com.example.demo_shai / app_flutter / flutter_assets / res_timestamp-1-1	-rw-----
> / com.example.demo_shai / app_flutter / cache	drwxrws--x
> / com.example.demo_shai / app_flutter / code_cache	drwxrws--x
> / com.example.demo_shai / app_flutter / files	drwxrwx--x
✓ / com.example.demo_shai / app_flutter / shared_prefs	drwxrwx--x
</> / com.example.demo_shai / app_flutter / shared_prefs / FlutterSharedPrefe	-rw-rw----

11. Open the file to observe its contents.

```

</> FlutterSharedPreferences.xml
1  <?xml version='1.0' encoding='utf-8' standalone="yes" type="text/plain">
2  <map>
3    <string name="flutter.name">Your name here</string>
4    <string name="flutter.email">Your email here</string>
5  </map>

```

Congratulations! Your app can now save data to the shared preference file.



TODO: Enhance the app to ensure all inputs are in the right format.

Name: ensure that name only contains letters, spaces, and potentially certain punctuation marks such as hyphens or apostrophes.

Email: A valid email address consists of an email prefix and an email domain, both in acceptable formats.